

Task 2: Task Management Application

Objective

Develop a **Task Management Web Application** that allows users to create, update, delete, and track tasks efficiently. The application should support user authentication and provide a responsive interface for managing daily activities.

Features Required

1. User Authentication & Authorization

Users should be able to:

- Register
- Login
- Logout

Security features:

- Password hashing using bcrypt
 - Session management
 - Protected routes
-

2. Task CRUD Operations

Create Task

Users can create tasks with:

- Title
- Description
- Due Date
- Priority

Example:

Title: Complete Internship Task
Description: Submit Task Management Project
Due Date: 25-06-2026
Priority: High

Read Tasks

View all tasks.

Update Tasks

Edit:

- Title
- Description
- Status
- Priority

Delete Tasks

Remove completed or unnecessary tasks.

3. Task Status Tracking

Each task should have:

Pending
In Progress
Completed

Example:

Task	Status
Build Login Page	Completed

Create Database	In Progress
Deploy Website	Pending

4. Responsive Design

Application should work on:

- Desktop
 - Tablet
 - Mobile
-

5. Optional Real-Time Updates

Advanced implementation:

- Flask-SocketIO
- WebSockets

Updates appear instantly without refreshing.

Recommended Technology Stack

Frontend

- HTML
- CSS
- Bootstrap 5
- JavaScript

Backend

- Flask

Database

- SQLite

Authentication

- Flask-Bcrypt
-

Database Design

Users Table

```
CREATE TABLE users(  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    username TEXT UNIQUE,  
    password TEXT  
);
```

Tasks Table

```
CREATE TABLE tasks(  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    title TEXT NOT NULL,  
    description TEXT,  
    due_date DATE,  
    priority TEXT,  
    status TEXT,  
    user_id INTEGER,  
    FOREIGN KEY(user_id) REFERENCES users(id)
```

);

Project Structure

task-management-app/

```
|
|
|— app.py
|— database.db
|— requirements.txt
|
|— templates/
|   |— login.html
|   |— register.html
|   |— dashboard.html
|   |— add_task.html
|   |— edit_task.html
|
|— static/
|   |— css/
|   |   |— style.css
|   |
|   |— js/
|       |— script.js
|
|— README.md
```

Main Functionalities

User Registration

Username

Password

Confirm Password

User Login

Username

Password

Dashboard

Display:

Total Tasks

Pending Tasks

Completed Tasks

Example:

Total Tasks: 10

Pending: 4

Completed: 6

Add New Task

Form fields:

Task Title

Description

Due Date

Priority

Edit Task

Users can:

- Change status
 - Update task details
-

Delete Task

Remove task permanently.

Sample Workflow

Step 1

Register Account

Username: Hemanth

Password: *****

Step 2

Login

Welcome Hemanth

Step 3

Create Task

Title: Build Portfolio

Priority: High

Status: Pending

Step 4

Update Status

Completed

Expected Output

Dashboard

Task Management Dashboard

Total Tasks: 5

1. Build Portfolio

Status: Completed

2. Internship Submission

Status: Pending

3. Learn Flask

Status: In Progress